Rishikesh Paudel

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Education

Bachelor in Computer Engineering **Apr 2022 – Apr 2026**

Institute of Engineering, Pulchowk Campus, Tribhuvan University, Lalitpur, Nepal
Aggregate: 81%

Publications and Manuscripts

1. **R. Paudel**, Y. Shrestha, P. R. Pokhrel, S. Parajuli, B. Joshi, S. Shrestha. "RTL Design and Latency Optimisation of an Interaction Network Accelerator on FPGA for Real-Time Particle Track Reconstruction." *Manuscript in preparation*, 2026.
2. **R. Paudel**, Y. Shrestha, P. R. Pokhrel, S. Parajuli, B. Joshi, S. Shrestha. "Sum Aggregation versus Scalar Edge-Conditioned Attention for TrackML Edge Classification." *Manuscript in preparation*, 2026.
3. **R. Paudel**, Y. Shrestha, P. R. Pokhrel. "Online Track Reconstruction Using GNN on FPGA for the ATLAS Experiment." CERN-OPEN-2026-007, CERN, Geneva, July 2026. cds.cern.ch/record/2966162

Research Experience

Bachelor's Thesis, Particle Track Reconstruction Using GNN on FPGA **May 2025 – Jul 2026**

Institute of Engineering, Pulchowk Campus, in collaboration with ATLAS researchers at CERN.
Supervised by Dr. Basanta Joshi (IOE), with Dr. Santosh Parajuli (UIUC) and Dr. Suyog Shrestha (Washington College).

- Collaborated with researchers in the CERN ATLAS experiment, as an external participant,
  - to design, train and compare Graph Attention Network (GAT) against Graph Interaction Network (IN) for particle trajectory reconstruction in High Luminosity Large Hadron Collider (HL-LHC).
  - to develop a low-latency AI accelerator on FPGA for real time particle tracking in the ATLAS event filter.
- Achieved 0.9892 recall with the edge-conditioned GAT against 0.9030 for the Interaction Network, which in turn reached 0.9705 precision against 0.8985. The Interaction Network was selected for hardware implementation, since attention's softmax is costly in fixed-point arithmetic.
- Achieved Verilog implementation of the Interaction Network at 8-bit fixed point, and applied various optimizations acheiving 8.42× faster implementation than the sequential baseline, at 86% chip logic utilisation and 14.7 W.

Research Projects

Deep Reinforcement Learning for Mapless Robot Navigation **2026**

SAC, PPO, Stable-Baselines3, ROS 2, Gazebo Harmonic

- Developed a reinforcement learning system for autonomous robot navigation, where the robot learns to reach target locations while avoiding obstacles. A Soft Actor-Critic (SAC) policy achieved 90% success in Gazebo across 20 random start/goal pairs, with a 5.3s mean time to reach the target, outperforming PPO (65%) and a proportional-controller baseline (40%).
- Identified and addressed a simulation transfer problem: a policy achieving 98.5% success in the training environment dropped to 5% when tested in Gazebo. Introduced domain randomization across movement, sensor, actuator and control-timing parameters, improving Gazebo success to 90% without changing the neural network or reward function.
- Built a lightweight 2-D training simulator that reproduced the Gazebo environment, LiDAR sensor inputs and robot control interface. Increased training throughput from 3 to 4,700 steps/s while improving Gazebo success from 60% to 90% compared with direct training in Gazebo.

Crop Disease Classification with Offline Mobile Deployment **2025**

ResNet-50, EfficientNet-B0, ViT, TensorFlow

- Trained and compared ResNet-50, EfficientNet-B0 and a fine-tuned Vision Transformer on a 34-class, 9-crop augmented plant disease dataset (120k train / 30k validation), reaching 99.85%, 99.58% and 98.34% validation accuracy respectively.
- Deployed a quantised model for fully offline mobile inference, motivated by the absence of reliable field connectivity for smallholder farmers in Nepal.

UAV-Based Rooftop Mapping and Segmentation for Terrace Farming **2025**

APM 2.8, U-Net, GIS

- Built a UAV from the ground up on an APM 2.8 flight controller with 6M GPS for aerial imaging.
- Achieved rooftop area estimation within 1.96% error on rectangular shapes, 3.90% on triangular and 6.17% on circular, from U-Net segmentation masks calibrated against a reference object, with masks exported as georeferenced GIS objects. Validation pixel accuracy 93.5% on a 25-image labelled set.

Professional Experience

AI/ML Engineer, MedForce AI, Remote (UK) **Aug 2026 – Present**

AI decision-simulation platform for the pharmaceutical research industry.

- Built synthetic prescriber personas using a retrieval-augmented LLM pipeline, allowing treatment decisions to be simulated across clinically distinct healthcare professional profiles.
- Developed an advisory board system that generates grounded market strategy recommendations, with every recommendation traceable to its retrieved supporting evidence.
- Applied TimesFM, a time-series foundation model, to forecast changes in drug utilisation following competitor entry into a market.

Chief Technology Officer, Wholyx LLC (FileCourt), Remote **Mar 2026 – Jun 2026**

Led engineering for a multi-tenant legal-tech SaaS platform for electronic court filing for U.S. law firms, managing a cumulative team of seven engineers. Guided the platform through from scratch to court certification.

Volunteering

Digital Literacy Instructor, LOCUS & Worldlink, Nepal **2025**

Taught digital literacy, cybersecurity, AI fundamentals, and networking to students in grades 8–12 across eight schools in the Karnali region and Kathmandu, developing curriculum materials for students in underserved rural communities.

Technical Skills

- **Machine Learning**: PyTorch, PyTorch Lightning, PyTorch Geometric, TensorFlow, scikit-learn, Graph Neural Networks, CNNs, U-Net, Vision Transformers, Reinforcement Learning
- **Efficient Inference**: Quantisation, fixed-point arithmetic, hardware-aware deployment, latency optimisation, offline mobile inference, performance benchmarking
- **Hardware**: Verilog, VHDL, RTL design, Vivado
- **Robotics and Sensing**: ROS2, Gazebo, 2D LiDAR, UAV assembly and flight testing, Mission Planner, GIS export and georeferencing
- **Programming**: Python, C, C++, SQL, JavaScript/TypeScript, Bash
- **Tooling**: Git, Docker, Kubernetes, Google Cloud, AWS, Ansible, Terraform

Honors and Awards

- First Place, 9th National FPGA Competition, Nepal, 2026
- First Place, 8th National FPGA Competition, Nepal, 2025
- Local Winner (Kathmandu), NASA Space Apps Challenge, 2024
- Category Winner, Best Use of AI, Sagarmatha Hacks, 2023
- Fusemachines AI Fellowship, 2025

Leadership

- Classroom Representative (48 students, four years), IOE Pulchowk Campus, Apr 2022 – Apr 2026
- Secretary, IOE IT Club, IOE Pulchowk Campus, Apr 2024 – May 2025